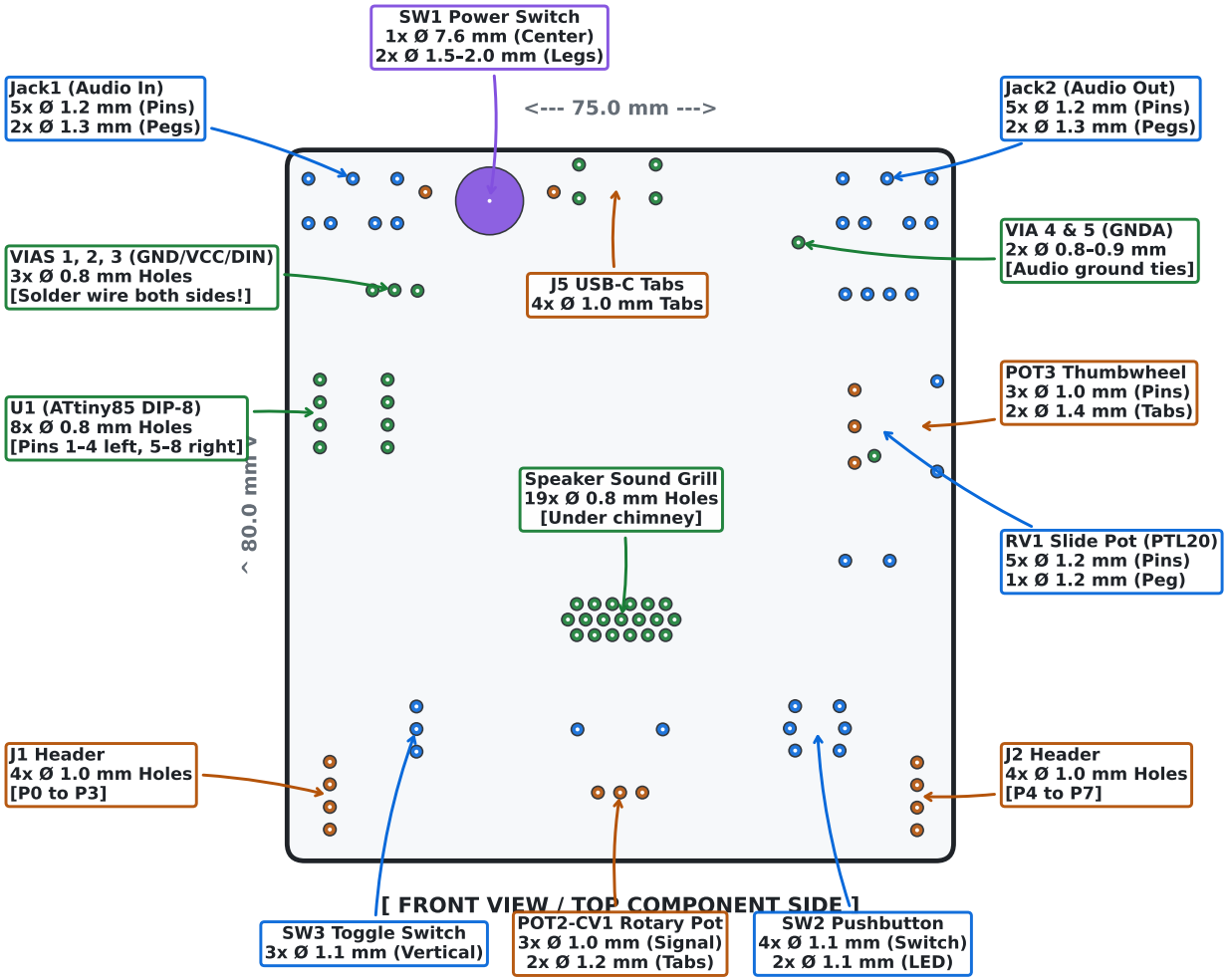


DRILL BIT SELECTION & CHECKLIST

Work from smallest to largest • 86 Holes Total

[ ] STEP 1: Ø 0.8 mm BIT (32 Holes)

- 19x Speaker Sound Grill (Acoustic center)
- 8x U1 ATtiny85 IC Socket (Pins 1–8)
- 5x Dedicated Vias (VIA 1, 2, 3, 4, 5)

[ ] STEP 2: Ø 1.0 mm BIT (18 Holes)

- 4x J1 Pin Socket Header (1x04)
- 4x J2 Pin Socket Header (1x04)
- 3x POT2-CV1 Rotary Pot (Signal pins)
- 3x POT3-Vol1 Thumbwheel (Signal pins)
- 4x J5 USB-C Mechanical Shield Tabs

[ ] STEP 3: Ø 1.2–1.3 mm BIT (33 Holes)

- 7x Jack1 (5 pins + 2 plastic pegs)
- 7x Jack2 (5 pins + 2 plastic pegs)
- 6x RV1 Slide Pot (5 pins + 1 peg)
- 6x SW2 Pushbutton KS01 (4 sw + 2 LED)
- 3x SW3 SPDT Sub-mini Toggle
- 4x Pot Side Bracket Tabs (POT2/POT3)

[ ] STEP 4: LARGE HOLES (SW1 Switch)

- 2x Ø 1.5–2.0 mm: SW1 power switch legs
- 1x Ø 7.5–8.0 mm: SW1 center button hole
  (Use step drill bit or round needle file)

DIY DRILLING SAFETY & WORKSHOP TIPS

1. Backing Board: Clamp flat scrap wood underneath to prevent copper delamination on exit.
2. High RPM / Low Pressure: Carbide micro-bits are brittle! Drill at 10,000+ RPM straight down with zero lateral tilt.
3. Pilot Centers: Your mask etched copper pilot marks, so 0.8 mm self-centers without bit wandering.
4. Bottom Side Drilling: If drilling from copper back, flip left-to-right (Jack1 is right, Jack2 is left).